ECON 0100 | Demo D

Solutions

*Answers and work are shown in the red blocks. Each skill earns **Pass** or **Not Yet**; the first red block under each skill says what a Pass shows.*

Hermione's Potion Ingredients

Hermione and Draco share a garden plot growing moonflowers for their potions project. Each can harvest 1 or 2 flowers weekly. More harvesting reduces future growth. The payoff matrix shows their utility from different harvesting choices.

		Draco: H1	Draco: H2
Hermione	Harvest 1 (H1)	10, 10	6, 15
	Harvest 2 (H2)	14, 6	8, 8

Outcomes are written (Hermione's choice, Draco's choice).

D1.1 | Best Responses

Pass (D1.1): Find all four best responses, H2 each time, by comparing a player's own payoffs with the other player's choice held fixed, and identify H2 as each player's dominant strategy.

	Hermione	**Draco**
Best Response When the Other Harvests 1:	**H2**	**H2**
Best Response When the Other Harvests 2:	**H2**	**H2**
Dominant Strategy:	☐ H1 ☒ **H2** ☐ none	☐ H1 ☒ **H2** ☐ none

Hold Draco's choice fixed and compare Hermione's payoffs, the first number in each cell, down his column. When Draco harvests 1, $14 > 10$, so she harvests 2. When Draco harvests 2, $8 > 6$, so she harvests 2.

Hold Hermione's choice fixed and compare Draco's payoffs, the second number, across her row. When Hermione harvests 1, $15 > 10$, so he harvests 2. When she harvests 2, $8 > 6$, so he harvests 2.

H2 is each player's best response to everything the other could do, so H2 is a dominant strategy for both. Best responses are boxed.

		Draco: H1	Draco: H2
Hermione	H1	10, 10	6, **15**
	H2	**14**, 6	**8**, **8**

D1.2 | Nash Equilibrium & Efficiency

Pass (D1.2): Check only (H2, H2) as the Nash equilibrium, identify (H1, H2) as efficient by comparing totals (16 at the equilibrium against 21), and identify (H1, H1) as the outcome both prefer to the equilibrium.

Nash Equilibrium (*check all that apply*):	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☒ **(H2, H2)**
Socially Efficient Outcome (*check all that apply*):	☐ (H1, H1)	☒ **(H1, H2)**	☐ (H2, H1)	☐ (H2, H2)
Total Payoff, Nash Equilibrium / Efficient Outcome:	**16** / **21**			
An Outcome Both Prefer to the Nash Equilibrium:	☒ **(H1, H1)**	☐ (H1, H2)	☐ (H2, H1)	☐ none

A Nash equilibrium is a cell where both payoffs are best responses, both boxed. Only (H2, H2) qualifies, so it is the one Nash equilibrium (yellow).

Efficiency compares total payoffs: (H1, H1) totals 20, (H1, H2) totals $6 + 15 = 21$, (H2, H1) totals $14 + 6 = 20$, and (H2, H2) totals 16. The efficient outcome is (H1, H2) (blue): Hermione harvests 1 and Draco harvests 2. Read the pair as (row, column); (H2, H1) totals only 20.

		Draco H1	**Draco** H2
Hermione	H1	10, 10	6, **15**
	H2	**14**, 6	**8**, **8**

The Nash equilibrium is not efficient: its total of 16 is 5 below 21. Both students would do better at (H1, H1), with 10 each instead of 8, but harvesting 2 is each one's dominant strategy, so neither holds back alone. This is a Prisoner's Dilemma. Neither (H1, H2) nor (H2, H1) is better for both: each leaves one student with 6, below the 8 of the equilibrium.

D2.1 | Common Resources

Pass (D2.1): Classify the moonflowers as rival and non-excludable, show that Hermione's switch gains her 2 while costing Draco 7, subtract the fine of 6 from every payoff for harvesting 2, and find the new equilibrium (H1, H1).

The Moonflowers in the Plot Are: ☒ **rival** ☐ non-rival ☐ excludable ☒ **non-excludable**

If Hermione Switches From (H1, H2) to (H2, H2): Hermione: **+2** Draco: **−7**

Professor Sprout begins checking the plot each week and fines anyone who harvests 2 flowers. The fine lowers that student's payoff by 6. Fill in the payoff matrix with the fine.

Nash Equilibrium With the Fine:

☒ **(H1, H1)** ☐ (H1, H2)

☐ (H2, H1) ☐ (H2, H2)

		Draco H1	**Draco** H2
Hermione	H1	**10, 10**	**6, 9**
	H2	**8, 6**	**2, 2**

The moonflowers are rival: a flower one student harvests is a flower the other can't, and harvesting more reduces future growth for both. They are non-excludable: the plot is shared, so neither can keep the other out. A rival, non-excludable good is a common resource.

Start at (H1, H2), where Draco harvests 2. If Hermione switches to 2, her payoff rises from 6 to 8, a gain of 2, but Draco's falls from 15 to 8, a loss of 7. Hermione counts her own gain and not Draco's loss, so she switches even though the total falls by 5. Draco reasons the same way, and the plot is overharvested.

The fine comes off Hermione's payoff in her H2 row and off Draco's payoff in his H2 column; at (H2, H2) both are fined. Now harvesting 1 is each student's best response to either choice: for Hermione $10 > 8$ and $6 > 2$, for Draco $10 > 9$ and $6 > 2$. The only Nash equilibrium is (H1, H1) (yellow), the sustainable harvest.

		Draco H1	**Draco** H2
Hermione	H1	**10**, **10**	**6**, 9
	H2	8, **6**	2, 2

When the other harvests 1, harvesting 2 is still worth 4 more to Hermione $(14 - 10)$ and 5 more to Draco $(15 - 10)$. The fine of 6 takes away more than that, so the rule makes the sustainable harvest each student's best response.

End-of-Term Fireworks

Five prefects are planning the end-of-term celebration. A fireworks show over the lake would cost 60 coins, and it can be seen from anywhere on the grounds. The table shows each prefect's benefit from the celebration (in coins), both without and with the fireworks.

Benefits (*in coins*)	**Hermione**	**Draco**	**Hannah**	**Cho**	**Padma**
Without fireworks	25	22	18	16	12
With fireworks	40	35	32	20	15

D3.1 | Public Goods

Pass (D3.1): Classify the fireworks show as non-rival and non-excludable, see that no prefect contributes voluntarily, find each prefect's marginal benefit as the rise in their benefit, add them to get a marginal social benefit of 49 coins, and compare it with the 60-coin cost.

The prefects first try to raise the 60 coins by splitting the cost into voluntary contributions of 12 coins. Each prefect decides in their own interest whether to contribute or not.

The Fireworks Show Is: ☐ rival ☒ **non-rival**
☐ excludable ☒ **non-excludable**

Prefects Who Would Contribute Voluntarily: **None**

	Hermione	**Draco**	**Hannah**	**Cho**	**Padma**
Marginal benefit of the fireworks	**15**	**13**	**14**	**4**	**3**

Marginal Social Benefit of the Fireworks: **49 coins**

Providing the Fireworks Is: ☐ efficient ☒ **not efficient**

The fireworks show is non-rival: one prefect watching takes nothing away from another's view. It is non-excludable: the show can be seen from anywhere on the grounds, so a prefect who pays nothing can't be kept from watching. A non-rival, non-excludable good is a public good.

No prefect contributes. Every prefect sees the show whether or not they pay, so each would rather keep their 12 coins and let the others pay. This is the free rider problem. Cho and Padma would not pay even if their 12 coins were the only way to get the show, since it is worth just 4 and 3 to them. Without them the 60 coins can't be raised, so a contribution from anyone else buys nothing.

Each prefect's marginal benefit from the fireworks is the rise in their benefit, with the fireworks minus without:

	Hermione	**Draco**	**Hannah**	**Cho**	**Padma**	**Total**
Without fireworks	25	22	18	16	12	93
With fireworks	40	35	32	20	15	142
Marginal benefit	15	13	14	4	3	49

$$MSB = 15 + 13 + 14 + 4 + 3 = 49 < 60 = MC$$

The show costs 11 coins more than it is worth to the prefects, so providing it is not efficient. Don't compare the with-fireworks total, 142, with the cost: the prefects get 93 of that from the celebration without the fireworks.

D3.2 | Voting on Public Goods

Pass (D3.2): Compare Hermione's 28 coins with the fireworks, after the fee, with her 25 without them to find her vote; count 3 for and 2 against; total 93 coins without and 82 with; and conclude that the fireworks pass but are not efficient.

Instead, the prefects put the fireworks to a vote. If a majority votes for them, each prefect pays a fee of 12 coins (60 coins total).

Hermione's Benefit Without Fireworks:	**25 coins**
Hermione's Benefit With Fireworks (*after the fee*):	**28 coins**
Hermione's Vote:	☒ **for** ☐ against
Prefects Voting For / Against:	**3** / **2**
Total Benefit Without Fireworks:	**93 coins**
Total Benefit With Fireworks (*after the fees*):	**82 coins**
Fireworks With a 12-Coin Fee:	☐ efficient ☒ **not efficient**

Each prefect votes for the fireworks if their benefit with them, after the 12-coin fee, is higher than their benefit without them. Hermione's is $40 - 12 = 28 > 25$, so she votes for them.

	Hermione	**Draco**	**Hannah**	**Cho**	**Padma**	**Total**
Without fireworks	25	22	18	16	12	93
With fireworks, after the fee	28	23	20	8	3	82
Vote	for	for	for	against	against	3 to 2

The fireworks pass 3 to 2, but total benefit falls from 93 to 82, so they are not efficient. The three prefects in favor gain $3 + 1 + 2 = 6$ coins, while Cho and Padma lose $8 + 9 = 17$. The vote makes Cho and Padma pay a 12-coin fee for a show worth only 4 and 3 to them. The 11-coin loss is the same gap as between the show's marginal social benefit, 49, and its 60-coin cost.